

ASHKAN SAMIMI

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ABOUT ME

I am an interdisciplinary scientist trained in physics and engineering, with expertise in microfluidics and photonics. During my PhD, I moved into microbiology and developed microfluidic technologies to address fundamental questions in microbial systems. I am particularly interested in bridging research and application to create a real-world impact.

PROFESSIONAL EXPERIENCE

Since 01/2025

Hans Knöll Institute, Jena, Germany
Scientific co-worker

- Incubating a startup concept
- Managing two scientific projects

03/2020 - 12/2024

Hans Knöll Institute, Jena, Germany
Doctoral researcher

- Developed a combinatorial sample preparation platform for droplet-based microbiological applications

EDUCATION

05/2020 -12/2024

Friedrich Schiller University, Jena, Germany
Doctoral student

09/2016 - 12/2019

Laser and Plasma Research Institute (LAPRI), Tehran, Iran
Master of Science in Photonics

09/2012 - 06/2016

University of Bonab, Bonab, Iran
Bachelor of Science in Laser and Optics Engineering

HONORS AND AWARDS

DAAD fellowship award (57600€)
Ranked top student in the graduate school program at LAPRI (GPA 18.25/20)
Ranked top student in the faculty of science and engineering at Bonab University (GPA 18.75/20)
Ranked 2nd among 1087 participants in the Iranian Nationwide entrance exam (Photonics)
Ranked 2nd in the first round of the Iranian Nationwide Physics Olympiad (2016)
Ranked 20th in the final round of the Iranian Nationwide Physics Olympiad (2016)
Ranked 4th in the first round of the Iranian Nationwide Physics Olympiad (2015)
Ranked 19th in the final round of the Iranian Nationwide Physics Olympiad (2015)

PUBLICATIONS (SELECTED)

1- **Samimi, A.** Verdon, N., Allen, R. Rosenbaum, M. A. **(2026)**. Probing antibiotic inhibition in small bacterial populations with combinatorial microfluidics. **Small Science**.
2- Ott, F., **Samimi, A.**, Hengoju, S., Meyer-Zedler, T., Schmitt, M., Rosenbaum, M. A., & Popp, J**(2026)**. Automated Imaging of Cultured Bacteria in Picoliter Droplets by Coherent Anti-Stokes Raman Scattering and Two-Photon Excited Fluorescence Microscopy**Journal of Raman Spectroscopy**.
3- Hengoju, S., Abdissa, K., Boto, S. T., **Samimi, A.**, Martin, K., Jacobsen, I. D., & Rosenbaum, M. A. **(2025)**. A droplet microfluidic strategy for cultivation, investigation, and high-throughput isolation of mouse gut microbiome bacteria. **Applied and Environmental Microbiology**.
4- **Samimi, A.**, Hengoju, S., Martin, K., & Rosenbaum, M. A. **(2025)**. Advancing droplet-based microbiological assays: optofluidic detection meets multiplexed droplet generation. **Analyst**.
5- **Samimi, A.**, Hengoju, S., & Rosenbaum, M. A. **(2024)**. Combinatorial Sample Preparation Platform for Droplet-Based Applications in Microbiology. **Sensors and Actuators B: Chemical**.
6- Graf, M., **Samimi, A.**, Rosenbaum, M. A., & Hengoju, S. **(2024)**. Tröpfchenmikrofluidik für das Enzymscreening. **BIOspektrum**.
7- Weber, T., Hengoju, S., **Samimi, A.**, Roth, M., Tovar, M. and Rosenbaum, M.A., **(2022)**. Recovery and isolation of individual microfluidic picoliter droplets by triggered deposition. **Sensors and Actuators B: Chemical**.

SKILLS AND RESEARCH INTERESTS

Research interests	Microfluidics and lab-on-a-chip technologies Automation in microbiology Technical innovations in microbiological research Microbial physiology Microbial biotechnology
Experimental	microfabrication techniques (e.g., UV-lithography, 3d printing), image processing (Python), data analysis (Python), optical sensors, general imaging techniques (e.g., fluorescence), general microbiology techniques, instrument control software development (LabVIEW)
Theoretical	Theoretical microfluidics, General microbiology
Computer languages	Python, LabVIEW (intermediate)
Simulation software	COMSOL Multiphysics: microfluidic module
Language	English: C1 German: B1 Persian: native